

71. Write marginal and conditional distribution for continuous Random Variables.

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For two jointly continuous random variables X and Y with a joint probability density function (pdf) $f(x, y)$:

1. Marginal Probability Density Functions:

The marginal pdf of X , denoted as $f_X(x)$ (or $g(x)$), is obtained by integrating the joint pdf over the entire range of Y :

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

The marginal pdf of Y , denoted as $f_Y(y)$ (or $h(y)$), is obtained by integrating the joint pdf over the entire range of X :

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

2. Conditional Probability Density Functions:

The conditional pdf of X given that $Y = y$ is defined as:

$$f(x|y) = \frac{f(x, y)}{f_Y(y)}, \quad \text{provided } f_Y(y) > 0$$

The conditional pdf of Y given that $X = x$ is defined as:

$$f(y|x) = \frac{f(x, y)}{f_X(x)}, \quad \text{provided } f_X(x) > 0$$

72. A continuous random variable X has a p.d.f. $f(x) = 3x^2$, $0 \leq x \leq 1$. Find a such that $P(X \leq a) = P(X > a)$.

Step 1: Understand the Probability Condition

The total probability over the range $[0, 1]$ is equal to 1.

Given $P(X \leq a) = P(X > a)$, and knowing that $P(X \leq a) + P(X > a) = 1$, we can deduce:

$$2P(X \leq a) = 1 \implies P(X \leq a) = \frac{1}{2}$$

Step 2: Integrate the PDF

We need to calculate the cumulative probability up to a :

$$P(X \leq a) = \int_0^a f(x) dx = \int_0^a 3x^2 dx$$

Step 3: Solve the Integral

$$\int_0^a 3x^2 dx = 3 \left[\frac{x^3}{3} \right]_0^a = [x^3]_0^a = a^3$$

Step 4: Equate to 0.5

$$a^3 = \frac{1}{2}$$

$$a = \left(\frac{1}{2} \right)^{1/3} \text{ or } \sqrt[3]{0.5}$$

$$a = \frac{1}{\sqrt[3]{2}} \approx 0.7937$$

73. State the Central Limit Theorem

Statement:

The Central Limit Theorem (CLT) states that if we take random samples of size n from any population (regardless of the population's distribution shape) with a mean μ and a finite variance σ^2 , the distribution of the sample means $\bar{X}$ will approach a Normal Distribution as the sample size n becomes large (usually $n \geq 30$).

Mathematically, the standardized variable Z converges to the standard normal distribution $N(0, 1)$ as $n \rightarrow \infty$:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow N(0, 1)$$

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74. Assume that the pair of dice is thrown and the random variable X is the sum of numbers that appears on two dice. Find the mean or the expectation of the random variable X .

****Step 1: Identify Sample Space and Random Variable X (Sum)**

When two dice are thrown, the total outcomes are $6 \times 6 = 36$.

The possible values for the sum X range from 2 ($1 + 1$) to 12 ($6 + 6$).

Step 2: Determine Probability $P(X)$ for each Sum

$X = 2: (1,1) \rightarrow 1/36$

$X = 3: (1,2), (2,1) \rightarrow 2/36$

$X = 4: (1,3), (2,2), (3,1) \rightarrow 3/36$

$X = 5: (1,4), (2,3), (3,2), (4,1) \rightarrow 4/36$

$X = 6: (1,5), (2,4), (3,3), (4,2), (5,1) \rightarrow 5/36$

$X = 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) \rightarrow 6/36$

$X = 8: (2,6), (3,5), (4,4), (5,3), (6,2) \rightarrow 5/36$

$X = 9: (3,6), (4,5), (5,4), (6,3) \rightarrow 4/36$

$X = 10: (4,6), (5,5), (6,4) \rightarrow 3/36$

$X = 11: (5,6), (6,5) \rightarrow 2/36$

$X = 12: (6,6) \rightarrow 1/36$

Step 3: Calculate Expectation $E[X] = \sum x \cdot P(x)$

$$E[X] = \frac{1}{36}[(2 \cdot 1) + (3 \cdot 2) + (4 \cdot 3) + (5 \cdot 4) + (6 \cdot 5) + (7 \cdot 6) + (8 \cdot 5) + (9 \cdot 4) + (10 \cdot 3) + (11 \cdot 2) + (12 \cdot 1)]$$

$$E[X] = \frac{1}{36}[2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12]$$

$$E[X] = \frac{252}{36} = 7$$

Answer: The mean (expectation) is 7.

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75. Find the probability distribution for the number of doublets in the three throw of a pair of dice.

Step 1: Identify Parameters

This is a Binomial Distribution problem.

Experiment: Throwing a pair of dice 3 times ($n = 3$).

Success: Getting a "doublet" (same number on both dice).

Doublets: $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

Number of successes: 6

Total outcomes in one throw: 36

Probability of success (p): $6/36 = 1/6$

Probability of failure (q): $1 - 1/6 = 5/6$

Random Variable X : Number of doublets (0, 1, 2, or 3).

Step 2: Calculate Probabilities using Binomial Formula

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

For $X = 0$ (No doublets):

$$P(0) = \binom{3}{0} \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = 1 \cdot 1 \cdot \frac{125}{216} = \frac{125}{216}$$

For $X = 1$ (One doublet):

$$P(1) = \binom{3}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = 3 \cdot \frac{1}{6} \cdot \frac{25}{36} = \frac{75}{216}$$

- For $X = 2$ (Two doublets):

$$P(2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 3 \cdot \frac{1}{36} \cdot \frac{5}{6} = \frac{15}{216}$$

- For $X = 3$ (Three doublets):

$$P(3) = \binom{3}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = 1 \cdot \frac{1}{216} \cdot 1 = \frac{1}{216}$$

Step 3: Write the Probability Distribution Table

X (Number of Doublets)	0	1	2	3
P(X)	$\frac{125}{216}$	$\frac{75}{216}$	$\frac{15}{216}$	$\frac{1}{216}$

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76. A continuous random variable X has a p.d.f. $f(x) = 3x^2$, $0 < x < 1$. Find $P(\frac{1}{3} \leq x \leq \frac{2}{3})$.

Step 1: Set up the Definite Integral

We need to integrate the probability density function $f(x)$ from lower limit $1/3$ to upper limit $2/3$.

$$P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right) = \int_{1/3}^{2/3} 3x^2 dx$$

Step 2: Solve the Integral

$$\int 3x^2 dx = x^3$$

Applying limits:

$$\left[x^3\right]_{1/3}^{2/3}$$

Step 3: Calculate Values

$$\begin{aligned} &= \left(\frac{2}{3}\right)^3 - \left(\frac{1}{3}\right)^3 \\ &= \frac{8}{27} - \frac{1}{27} \\ &= \frac{7}{27} \end{aligned}$$

Answer: $\frac{7}{27} \approx 0.259$

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77. The joint probability distribution of two random variables X and Y is given by: $P(X = 0, Y = 1)$, $P(X = 1, Y = -1) = 1/3$ and $P(X = 1, Y = 1) = 1/3$. Find marginal distributions of X and Y , and the conditional probability distribution of X given $Y = 1$.

Step 1: Determine the missing probability.

The sum of all joint probabilities must equal 1.

Given:

- $P(1, -1) = 1/3$

- $P(1, 1) = 1/3$
- Let $P(0, 1) = p$

Sum: $p + 1/3 + 1/3 = 1 \implies p = 1/3$.

So, $P(X = 0, Y = 1) = 1/3$.

Step 2: Construct the Joint Probability Table.

$X \setminus Y$	-1	1	Marginal P(X)
0	0	1/3	1/3
1	1/3	1/3	2/3
Marginal P(Y)	1/3	2/3	1

Step 3: Marginal Distributions.

- **Marginal Distribution of X:**
 - $P(X = 0) = 1/3$
 - $P(X = 1) = 2/3$
- **Marginal Distribution of Y:**
 - $P(Y = -1) = 1/3$
 - $P(Y = 1) = 2/3$

Step 4: Conditional Probability Distribution of X given $Y = 1$.

Formula: $P(X = x|Y = 1) = \frac{P(X=x, Y=1)}{P(Y=1)}$

- For $X = 0$:

$$P(X = 0|Y = 1) = \frac{1/3}{2/3} = \frac{1}{2}$$

- For $X = 1$:

$$P(X = 1|Y = 1) = \frac{1/3}{2/3} = \frac{1}{2}$$

78. A petrol pump is supplied with petrol once a day. If its daily volume of sales (x) in thousands of liters is distributed by $f(x) = 5(1 - x)^4$, $0 \leq x \leq 1$, what must be the capacity of its tank in order that the probability that its supply will be exhausted in a given day shall be 0.01?

Step 1: Define the condition.

Let c be the tank capacity. The supply is exhausted if sales X exceed capacity c .

We need to find c such that $P(X > c) = 0.01$.

Step 2: Set up the integral.

$$P(X > c) = \int_c^1 f(x) dx = 0.01$$

$$\int_c^1 5(1 - x)^4 dx = 0.01$$

Step 3: Solve the integral.

Let $u = 1 - x$, then $du = -dx$.

When $x = c$, $u = 1 - c$. When $x = 1$, $u = 0$.

Integral becomes:

$$\begin{aligned} \int_{1-c}^0 5u^4(-du) &= \int_0^{1-c} 5u^4 du \\ &= [u^5]_0^{1-c} = (1 - c)^5 \end{aligned}$$

Step 4: Solve for c .

$$(1 - c)^5 = 0.01$$

$$1 - c = (0.01)^{1/5}$$

$$1 - c \approx 0.398$$

$$c \approx 1 - 0.398 = 0.602$$

Answer: The capacity must be approximately **602 liters** (since unit is thousands of liters, 0.602×1000).

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79.If a random variable X has density function $f(x) = 1/4$ for $-2 < x < 2$ (and 0 elsewhere). Obtain i. $P(|X|>1)$ and ii. $P(2X+3>5)$

1. Find $P(|X|>1)$

The inequality $|X|>1$ means the distance from 0 is greater than 1. This splits into two regions:

1. $X < -1$
2. $X > 1$

We need to integrate the probability density function $f(x)=1/4$ over the valid range of X (which is between -2 and 2).

Step 1: Define the limits

- For $X < -1$: The valid range within the domain is $-2 < x < -1$.
- For $X > 1$: The valid range within the domain is $1 < x < 2$.

Step 2: Set up the integrals

$$P(|X|>1)=P(-2<X<-1)+P(1<X<2)$$

$$P(|X|>1)=\int_{-2}^{-1} \frac{1}{4} dx + \int_1^2 \frac{1}{4} dx$$

Step 3: Solve the integrals

$$=\frac{1}{4}[x]_{-2}^{-1} + \frac{1}{4}[x]_1^2$$

$$=\frac{1}{4}(-1-(-2)) + \frac{1}{4}(2-1)$$

$$=41(1)+41(1)$$

$$=41+41=42=0.5$$

Answer: $P(|X|>1)=0.5$



2. Find $P(2X+3>5)$

Step 1: Simplify the inequality We need to isolate X:

$$2X+3>5$$

$$2X>5-3$$

$$2X>2$$

$$X>1$$

Step 2: Define the limits We need to find the probability that $X>1$. Given the domain of X is $-2<x<2$, the valid region for this condition is $1<x<2$.

Step 3: Set up the integral

$$P(X>1)=\int_1^2 f(x)dx$$

$$P(X>1)=\int_1^2 41dx$$

Step 4: Solve the integral

$$=41[x]_1^2$$

$$=41(2-1)$$

$$=41(1)=0.25$$

Answer: $P(2X+3>5)=0.25$



80. If X and Y are two random variables having the joint density function:

$$f(x, y) = \begin{cases} \frac{1}{8}(6 - x - y) & 0 \leq x < 2, 2 \leq y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find: i. $P(X < 1 \cap Y < 3)$, ii. $P(X + Y < 3)$, iii. $P(X < 1 | Y < 3)$

Solution:

i. Find $P(X < 1 \cap Y < 3)$

This is the integral over the rectangle $0 \leq x < 1$ and $2 \leq y < 3$.

$$\begin{aligned} P &= \int_2^3 \int_0^1 \frac{1}{8}(6 - x - y) dx dy \\ &= \frac{1}{8} \int_2^3 \left[6x - \frac{x^2}{2} - xy \right]_0^1 dy \\ &= \frac{1}{8} \int_2^3 (6 - 0.5 - y) dy = \frac{1}{8} \int_2^3 (5.5 - y) dy \\ &= \frac{1}{8} \left[5.5y - \frac{y^2}{2} \right]_2^3 \\ &= \frac{1}{8} [(16.5 - 4.5) - (11 - 2)] = \frac{1}{8} [12 - 9] = \frac{3}{8} \end{aligned}$$

ii. Find $P(X + Y < 3)$

Region defined by $x \geq 0$, $y \geq 2$ and $x + y < 3$ (which means $x < 3 - y$).

Since y must be at least 2, and x must be positive, y can range from 2 to 3. (If $y > 3$, x would be negative).

$$P = \int_2^3 \int_0^{3-y} \frac{1}{8}(6 - x - y) dx dy$$

Inner integral w.r.t x :

$$\left[6x - \frac{x^2}{2} - xy \right]_0^{3-y} = 6(3 - y) - \frac{(3 - y)^2}{2} - y(3 - y)$$

$$= (3 - y) \left[6 - \frac{3 - y}{2} - y \right] = (3 - y) \left[\frac{12 - 3 + y - 2y}{2} \right] = \frac{1}{2} (3 - y)(9 - y)$$

Now integrate w.r.t y from 2 to 3:

$$\begin{aligned} \frac{1}{16} \int_2^3 (27 - 3y - 9y + y^2) dy &= \frac{1}{16} \int_2^3 (y^2 - 12y + 27) dy \\ &= \frac{1}{16} \left[\frac{y^3}{3} - 6y^2 + 27y \right]_2^3 \end{aligned}$$

Values at 3: $(9 - 54 + 81) = 36$

Values at 2: $(8/3 - 24 + 54) = 32.66... = 98/3$

$$\text{Result: } \frac{1}{16} (36 - 32.66) = \frac{1}{16} \left(\frac{108}{3} - \frac{98}{3} \right) = \frac{1}{16} \cdot \frac{10}{3} = \frac{5}{24}$$

iii. Find $P(X < 1 | Y < 3)$

$$P(X < 1 | Y < 3) = \frac{P(X < 1 \cap Y < 3)}{P(Y < 3)}$$

We found the numerator in part (i): $3/8$.

We need $P(Y < 3)$.

$$P(Y < 3) = \int_2^3 \int_0^2 \frac{1}{8} (6 - x - y) dx dy$$

Inner integral (0 to 2):

$$\left[6x - \frac{x^2}{2} - xy \right]_0^2 = 12 - 2 - 2y = 10 - 2y$$

Outer integral (2 to 3):

$$\begin{aligned} \frac{1}{8} \int_2^3 (10 - 2y) dy &= \frac{1}{8} [10y - y^2]_2^3 \\ &= \frac{1}{8} [(30 - 9) - (20 - 4)] = \frac{1}{8} [21 - 16] = \frac{5}{8} \end{aligned}$$

Calculation:

$$\frac{3/8}{5/8} = \frac{3}{5}$$

81. Five defective bulbs are accidentally mixed with twenty good ones. Find the probability distribution of the number of defective bulbs, if four bulbs are drawn at random.

Step 1: Identify Parameters.

- Total bulbs = 25
- Defective (D) = 5
- Good (G) = 20
- Draws (n) = 4
- Let X = number of defective bulbs drawn (0, 1, 2, 3, 4).
- Total ways to draw 4 from 25: $N = \binom{25}{4} = \frac{25 \cdot 24 \cdot 23 \cdot 22}{4 \cdot 3 \cdot 2 \cdot 1} = 12,650$.

Step 2: Calculate Probability for each X .

Formula: $P(X = x) = \frac{\binom{5}{x} \binom{20}{4-x}}{12650}$

- $X = 0$:

$$P(0) = \frac{\binom{5}{0} \binom{20}{4}}{12650} = \frac{1 \cdot 4845}{12650} \approx 0.383$$

- $X = 1$:

$$P(1) = \frac{\binom{5}{1} \binom{20}{3}}{12650} = \frac{5 \cdot 1140}{12650} = \frac{5700}{12650} \approx 0.451$$

- $X = 2$:

$$P(2) = \frac{\binom{5}{2} \binom{20}{2}}{12650} = \frac{10 \cdot 190}{12650} = \frac{1900}{12650} \approx 0.150$$

- $X = 3$:

$$P(3) = \frac{\binom{5}{3} \binom{20}{1}}{12650} = \frac{10 \cdot 20}{12650} = \frac{200}{12650} \approx 0.016$$

- $X = 4$:

$$P(4) = \frac{\binom{5}{4} \binom{20}{0}}{12650} = \frac{5 \cdot 1}{12650} \approx 0.0004$$

Distribution Table:

X	0	1	2	3	4
$P(X)$	$\frac{4845}{12650}$	$\frac{5700}{12650}$	$\frac{1900}{12650}$	$\frac{200}{12650}$	$\frac{5}{12650}$

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82. If X and Y are two random variables having the joint probability mass function $p(x, y) = \frac{1}{27}(2x + y)$ for $x, y = 0, 1, 2$. Find the conditional distribution of Y for $X = x$.

Solution:

Step 1: Find the Marginal Distribution of X , $P_X(x)$.

$$P_X(x) = \sum_{y=0}^2 p(x, y)$$

$$P_X(x) = \frac{1}{27}[(2x + 0) + (2x + 1) + (2x + 2)]$$

$$P_X(x) = \frac{1}{27}[6x + 3] = \frac{3(2x + 1)}{27} = \frac{2x + 1}{9}$$

Step 2: Find Conditional Distribution $P(Y = y|X = x)$.

$$P(y|x) = \frac{p(x, y)}{P_X(x)}$$

$$P(y|x) = \frac{\frac{1}{27}(2x + y)}{\frac{2x+1}{9}}$$

$$P(y|x) = \frac{2x+y}{27} \cdot \frac{9}{2x+1}$$

$$P(y|x) = \frac{2x+y}{3(2x+1)}$$

Answer: The conditional distribution is $P(Y = y|X = x) = \frac{2x+y}{3(2x+1)}$ for $y = 0, 1, 2$.

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83. If the pdf of continuous random variable X is $f(x) = kx^2(1-x)$ for $0 < x < 1$. Calculate the value of k and mean for X .

Solution:

Step 1: Find k .

Total probability must be 1.

$$\int_0^1 kx^2(1-x) dx = 1$$

$$k \int_0^1 (x^2 - x^3) dx = 1$$

$$k \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 1$$

$$k \left(\frac{1}{3} - \frac{1}{4} \right) = 1 \implies k \left(\frac{1}{12} \right) = 1$$

$$k = 12$$

Step 2: Find the Mean $E[X]$.

$$E[X] = \int_0^1 x \cdot f(x) dx$$

$$E[X] = \int_0^1 x \cdot 12x^2(1-x) dx$$

$$E[X] = 12 \int_0^1 (x^3 - x^4) dx$$

$$E[X] = 12 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1$$

$$E[X] = 12 \left(\frac{1}{4} - \frac{1}{5} \right) = 12 \left(\frac{5-4}{20} \right) = 12 \left(\frac{1}{20} \right)$$

$$E[X] = \frac{12}{20} = \frac{3}{5}$$

Answer: $k = 12$, Mean = 0.6

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84 Suppose the p.d.f of a continuous random variable X is defined as

$$f(x) = \begin{cases} 1+x, & -1 < x < 0 \\ 1-x, & 0 \leq x < 1 \end{cases}$$

Find the cumulative distribution function (c.d.f) $F(x)$.

Solution:

The Cumulative Distribution Function (CDF) is defined as $F(x) = \int_{-\infty}^x f(t) dt$. We calculate this for each interval.

1. For $x \leq -1$:

The probability density is 0.

$$F(x) = 0$$

2. For $-1 < x \leq 0$:

We integrate $1+t$ from -1 to x .

$$\begin{aligned}
 F(x) &= \int_{-1}^x (1+t) dt = \left[t + \frac{t^2}{2} \right]_{-1}^x \\
 &= \left(x + \frac{x^2}{2} \right) - \left(-1 + \frac{1}{2} \right) = x + \frac{x^2}{2} + \frac{1}{2} \\
 &= \frac{1}{2}(x^2 + 2x + 1) = \frac{1}{2}(x+1)^2
 \end{aligned}$$

3. For $0 < x < 1$:

We add the full probability from the previous interval ($F(0)$) plus the integral of the current interval.

From step 2, $F(0) = \frac{1}{2}(0+1)^2 = 0.5$.

$$\begin{aligned}
 F(x) &= 0.5 + \int_0^x (1-t) dt \\
 &= 0.5 + \left[t - \frac{t^2}{2} \right]_0^x = 0.5 + x - \frac{x^2}{2}
 \end{aligned}$$

4. For $x \geq 1$:

The total probability is accumulated.

$$F(x) = 1$$

Answer:

$$F(x) = \begin{cases} 0 & x \leq -1 \\ \frac{1}{2}(x+1)^2 & -1 < x \leq 0 \\ \frac{1}{2} + x - \frac{x^2}{2} & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$

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85 A random variable X has the following probability mass function (values involving c).

X	0	1	2
P(X)	$3c^3$	$4c - 10c^2$	$5c - 1$

Tasks:

i. Find c .

ii. Evaluate $P(X < 2)$, $P(2X + 3 \geq 5)$, and $P(1 < X \leq 2)$.

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Step 1: Find the value of c

For a valid probability distribution, the sum of all probabilities must equal 1.

$$\sum P(X) = 1$$

$$P(0) + P(1) + P(2) = 1$$

$$3c^3 + (4c - 10c^2) + (5c - 1) = 1$$

Rearrange the terms into a standard cubic equation form:

$$3c^3 - 10c^2 + 9c - 1 = 1$$

$$3c^3 - 10c^2 + 9c - 2 = 0$$

Solve for c :

We look for integer or rational roots. Let's test $c = 1$:

$$3(1)^3 - 10(1)^2 + 9(1) - 2 = 3 - 10 + 9 - 2 = 0$$

Since $c = 1$ is a root, $(c - 1)$ is a factor.

Now, we factor the cubic equation $(3c^3 - 10c^2 + 9c - 2)$ by dividing by $(c - 1)$:

The resulting quadratic is $3c^2 - 7c + 2$.

So, $(c - 1)(3c^2 - 7c + 2) = 0$.

Factor the quadratic $3c^2 - 7c + 2$:

$$(3c - 1)(c - 2) = 0$$

The possible values for c are 1, 2, and $\frac{1}{3}$.

Validate c :

Probabilities $P(X)$ must be between 0 and 1.

- If $c = 1$: $P(2) = 5(1) - 1 = 4$ (Invalid, probability > 1).
- If $c = 2$: $P(2) = 5(2) - 1 = 9$ (Invalid).
- If $c = 1/3$:
 - $P(0) = 3(1/3)^3 = 3(1/27) = 1/9$
 - $P(1) = 4(1/3) - 10(1/3)^2 = 4/3 - 10/9 = 12/9 - 10/9 = 2/9$
 - $P(2) = 5(1/3) - 1 = 5/3 - 3/3 = 2/3$ (which is $6/9$)
 - **Sum:** $1/9 + 2/9 + 6/9 = 9/9 = 1$. (Valid)

Therefore, $c = \frac{1}{3}$.

The Probability Distribution Table is:

X	0	1	2
$P(X)$	1/9	2/9	6/9 (or 2/3)

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Step 2: Evaluate the Probabilities

A. Find $P(X < 2)$

This includes all values of X strictly less than 2 ($X = 0$ and $X = 1$).

$$P(X < 2) = P(0) + P(1)$$

$$P(X < 2) = \frac{1}{9} + \frac{2}{9} = \frac{3}{9}$$

Answer: $\frac{1}{3}$

B. Find $P(2X + 3 \geq 5)$

First, solve the inequality for X :

$$2X \geq 5 - 3$$

$$2X \geq 2$$

$$X \geq 1$$

This includes values $X = 1$ and $X = 2$.

$$P(X \geq 1) = P(1) + P(2)$$

$$P(X \geq 1) = \frac{2}{9} + \frac{6}{9} = \frac{8}{9}$$

Answer: $\frac{8}{9}$

C. Find $P(1 < X \leq 2)$

This requires X to be strictly greater than 1 and less than or equal to 2. The only integer value in the table that satisfies this is $X = 2$.

$$P(1 < X \leq 2) = P(2)$$

$$P(2) = \frac{6}{9} = \frac{2}{3}$$

Answer: $\frac{2}{3}$

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86. A random variable X is distributed at random between the values 0 and 1. Its probability density function is $f(x) = kx^2(1 - x^3)$. Find the value of k , find its mean and variance.

Solution:

1. Find k :

$$\int_0^1 kx^2(1 - x^3) dx = 1$$

$$k \int_0^1 (x^2 - x^5) dx = 1$$

$$k \left[\frac{x^3}{3} - \frac{x^6}{6} \right]_0^1 = 1$$

$$k \left(\frac{1}{3} - \frac{1}{6} \right) = 1 \implies k \left(\frac{1}{6} \right) = 1$$

$$k = 6$$

2. Find Mean (μ):

$$E[X] = \int_0^1 x \cdot f(x) dx = \int_0^1 6x(x^2 - x^5) dx$$

$$= 6 \int_0^1 (x^3 - x^6) dx$$

$$= 6 \left[\frac{x^4}{4} - \frac{x^7}{7} \right]_0^1 = 6 \left(\frac{1}{4} - \frac{1}{7} \right)$$

$$= 6 \left(\frac{7 - 4}{28} \right) = 6 \left(\frac{3}{28} \right) = \frac{18}{28} = \frac{9}{14}$$

Mean $\mu \approx 0.643$

3. Find Variance (σ^2):

First, find $E[X^2]$:

$$E[X^2] = \int_0^1 x^2 \cdot f(x) dx = \int_0^1 6x^2(x^2 - x^5) dx$$

$$= 6 \int_0^1 (x^4 - x^7) dx$$

$$= 6 \left[\frac{x^5}{5} - \frac{x^8}{8} \right]_0^1 = 6 \left(\frac{1}{5} - \frac{1}{8} \right)$$

$$= 6 \left(\frac{8-5}{40} \right) = 6 \left(\frac{3}{40} \right) = \frac{18}{40} = \frac{9}{20}$$

Variance formula: $Var(X) = E[X^2] - (E[X])^2$

$$\begin{aligned} Var(X) &= \frac{9}{20} - \left(\frac{9}{40} \right)^2 \\ &= 0.45 - \frac{81}{196} \approx 0.45 - 0.4132 \\ Var(X) &\approx 0.0368 \end{aligned}$$

(In fraction: 9/245)

✍

87. Let X and Y be jointly distributed with pdf $f(x, y) = \frac{1}{4}(1 + xy)$ for $|x| < 1, |y| < 1$. Show that X and Y are not independent but X^2 and Y^2 are independent.

1. Show X and Y are Dependent:

For independence, $f(x, y)$ must equal $f_X(x) \cdot f_Y(y)$.

Find marginal $f_X(x)$:

$$\begin{aligned} f_X(x) &= \int_{-1}^1 \frac{1}{4}(1 + xy) dy = \frac{1}{4} \left[y + \frac{xy^2}{2} \right]_{-1}^1 \\ &= \frac{1}{4} [(1 + x/2) - (-1 + x/2)] = \frac{1}{4}(2) = \frac{1}{2} \end{aligned}$$

Similarly, $f_Y(y) = 1/2$.

Product of marginals: $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Since $\frac{1}{4} \neq \frac{1}{4}(1 + xy)$ (unless $xy = 0$), X and Y are NOT independent.

2. Show X^2 and Y^2 are Independent:

Consider the joint cumulative distribution of $U = X^2$ and $V = Y^2$:

$$\begin{aligned} P(X^2 \leq u, Y^2 \leq v) &= P(-\sqrt{u} < X < \sqrt{u}, -\sqrt{v} < Y < \sqrt{v}) \\ &= \int_{-\sqrt{v}}^{\sqrt{v}} \int_{-\sqrt{u}}^{\sqrt{u}} \frac{1}{4}(1 + xy) dx dy \\ &= \frac{1}{4} \left(\int \int 1 dx dy + \int \int xy dx dy \right) \end{aligned}$$

The integral of xy over a region symmetric about the origin is 0.

$$\begin{aligned} &= \frac{1}{4} \int_{-\sqrt{v}}^{\sqrt{v}} \int_{-\sqrt{u}}^{\sqrt{u}} 1 dx dy \\ &= \frac{1}{4} (2\sqrt{u})(2\sqrt{v}) = \sqrt{u}\sqrt{v} \end{aligned}$$

Since the joint probability $F(u, v) = \sqrt{u}\sqrt{v}$ factors into a function of u only ($\sqrt{u}$) and v only ($\sqrt{v}$), X^2 and Y^2 are independent.

⌘

88. The joint probability density function of (X, Y) is $f(x, y) = 2$ for $0 \leq x < 1, 0 < y < x$.

Solution:

i. Marginal density functions:

- X (Integrate out y): The limits for y are 0 to x .

$$f_X(x) = \int_0^x 2 dy = 2[y]_0^x = 2x, \quad 0 \leq x < 1$$

- Y (Integrate out x): Since $y < x < 1$, the limits for x are y to 1.

$$f_Y(y) = \int_y^1 2 dx = 2[x]_y^1 = 2(1 - y), \quad 0 < y < 1$$

ii. Conditional density functions:

- $f(y|x)$:

$$f(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x}, \quad 0 < y < x$$

- $f(x|y)$:

$$f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2}{2(1-y)} = \frac{1}{1-y}, \quad y < x < 1$$

∞

89. A random variable X has a given probability distribution involving k .

X	0	1	2	3	4	5	6	7
P(x)	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Solution:

i. Find k :

Sum of probabilities must equal 1.

$$\sum P(x) = k + 2k + 2k + 3k + k^2 + 2k^2 + (7k^2 + k) = 1$$

$$10k^2 + 9k = 1 \implies 10k^2 + 9k - 1 = 0$$

Factor the quadratic: $(10k - 1)(k + 1) = 0$.

Possible roots: $k = 0.1$ or $k = -1$. Since probability cannot be negative, $k = 0.1$.

ii. Evaluate:

- $P(X < 6)$: Sum of probabilities from $X = 0$ to $X = 5$. Alternatively, $1 - P(X \geq 6)$.

$$P(X \geq 6) = P(6) + P(7) = 2k^2 + (7k^2 + k) = 9k^2 + k$$

Using $k = 0.1$: $9(0.01) + 0.1 = 0.09 + 0.1 = 0.19$.

$$P(X < 6) = 1 - 0.19 = 0.81$$

- $P(3 < X \leq 6)$: Includes $X = 4, 5, 6$.

$$\begin{aligned} P(4) + P(5) + P(6) &= 3k + k^2 + 2k^2 = 3k + 3k^2 \\ &= 3(0.1) + 3(0.01) = 0.3 + 0.03 = 0.33 \end{aligned}$$

iii. Find minimum x so that $P(X \leq x) > 1/2$:

Calculate Cumulative Probabilities (CDF):

- $P(X \leq 0) = 0$
- $P(X \leq 1) = 0.1$
- $P(X \leq 2) = 0.1 + 0.2 = 0.3$
- $P(X \leq 3) = 0.3 + 0.2 = 0.5$ (Exactly $1/2$)
- $P(X \leq 4) = 0.5 + 0.3 = 0.8$ (Greater than $1/2$)

The minimum x where the cumulative sum is strictly *greater* than $1/2$ is $x = 4$.

↻

90. Joint distribution of X and Y is given by

$f(x, y) = 4xye^{-(x^2+y^2)}$; $x \geq 0, y \geq 0$. Test whether X and Y are independent. Find the conditional density of X given $Y = y$.

****1. Test for Independence:**

Can $f(x, y)$ be written as $g(x)h(y)$?

$$f(x, y) = 4xye^{-x^2}e^{-y^2} = (2xe^{-x^2}) \cdot (2ye^{-y^2})$$

Let $g(x) = 2xe^{-x^2}$ and $h(y) = 2ye^{-y^2}$.

Since the joint PDF factors perfectly into a function of x and a function of y , and the limits ($x \geq 0, y \geq 0$) do not depend on each other:

Yes, X and Y are Independent.

****2. Find Conditional Density $f(x|y)$:**

Formula: $f(x|y) = \frac{f(x,y)}{f_Y(y)}$.

Since X and Y are independent, the conditional density of X given Y is simply the marginal density of X .

$$f(x|y) = f_X(x) = 2xe^{-x^2}, \quad x \geq 0$$